Power Car Seat

Operation Manual

Abstract

This document provides comprehensive operational instructions for the Panthera car seat control system. It details procedures for operating the system through physical buttons, UART communication, and CAN communication. This manual serves as a reference for users and technicians who need to interact with the system, ensuring proper usage and maximizing functionality.

Revision Note

Version	Date	Author	Description of Changes
v0.0.1	2025-10-01	Uy Nguyen Quoc	Initial document

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1. Introduction

The Panthera Power Car Seat system provides multiple methods of operation to accommodate different use cases and integration scenarios. This manual covers three primary methods of interaction:

1. **Button Operation** - Using the physical control buttons on the seat or control panel
2. **UART Communication** - Interfacing with the system via serial communication
3. **CAN Communication** - Controlling the system through automotive CAN bus networks

Each method provides complete access to the system's functionality with varying degrees of complexity and integration requirements. This document provides step-by-step instructions for all operational modes to ensure proper system usage.

2. Button Operation Instructions

2.1 Overview and Button Functions

The Panthera seat control system features a set of intuitive buttons for controlling various seat positions and memory functions. The buttons are strategically placed for easy access and operation.

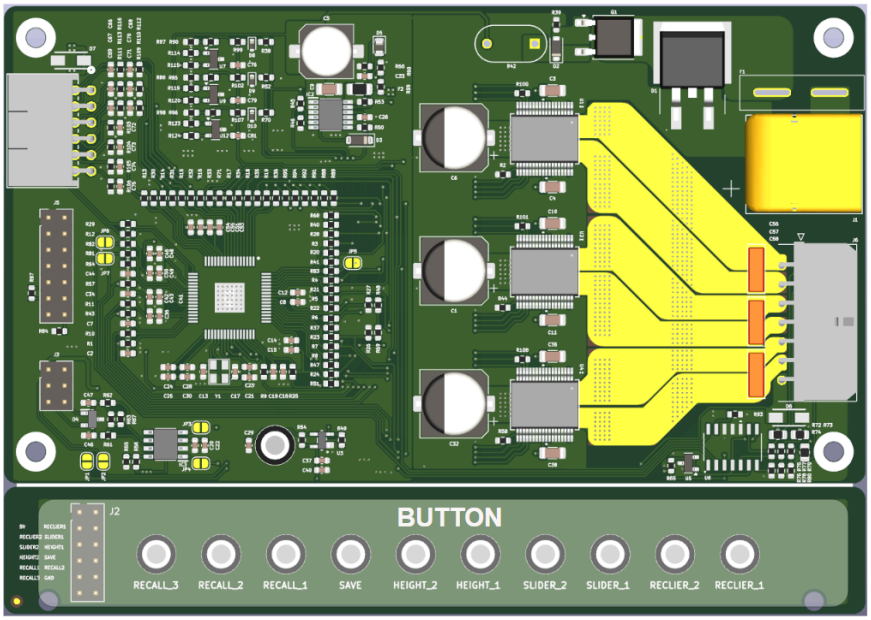


Figure 2.1: Button Layout of Panthera Seat Control System (Top view)

The table below provides a comprehensive overview of all buttons, their functions, and associated actions:

Table 2.1: Button Functions in the Panthera Seat Control System

No.	Button Name	Function	Action
1	RECLINER_1	Seatback Upright Position	Press and hold to return seatback to upright position. Release to stop movement.
2	RECLINER_2	Seatback Recline Position	Press and hold to recline seatback. Release to stop movement.
3	SLIDER_1	Seat Forward Adjustment	Press and hold to move the seat forward. Release to stop

No.	Button Name	Function	Action
			movement.
4	SLIDER_2	Seat Backward Adjustment	Press and hold to move the seat backward. Release to stop movement.
5	HEIGHT_1	Seat Upward Movement	Press and hold to raise the seat. Release to stop movement.
6	HEIGHT_2	Seat Downward Movement	Press and hold to lower the seat. Release to stop movement.
7	SAVE	Save Current Position	Press to begin the position saving sequence.
8	RECALL_1	Memory Position 1	Press and short hold to recall saved position 1 or save current position to memory slot 1.
9	RECALL_2	Memory Position 2	Press and short hold to recall saved position 2 or save current position to memory slot 2.
10	RECALL_3	Memory Position 3	Press and short hold to recall saved position 3 or save current position to memory slot 3.

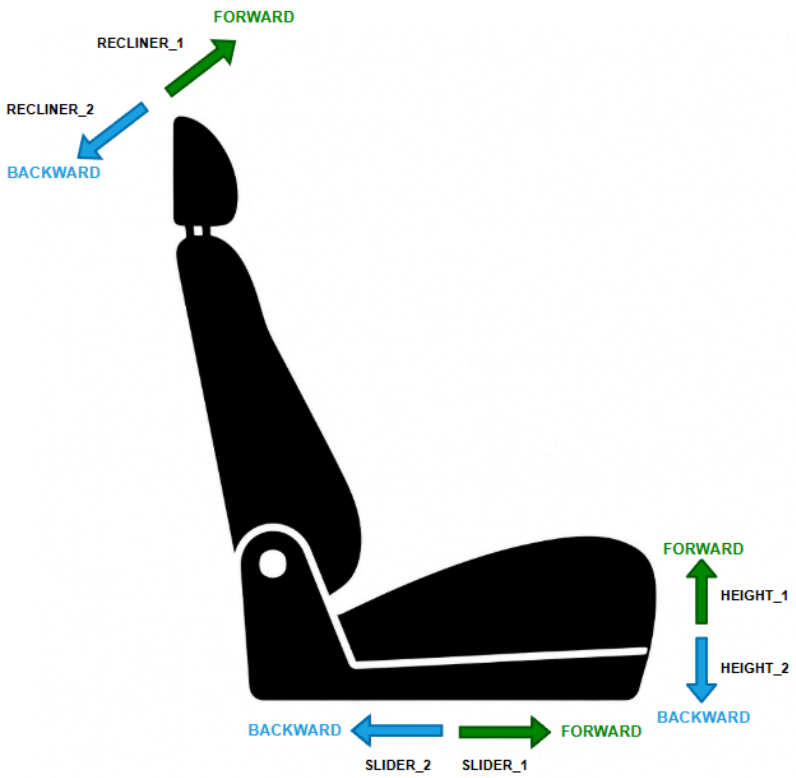


Figure 2.2: Seat Movement Directions Corresponding to Buttons

2.2 Operation Modes

The Panthera seat control system offers three primary operation modes:

1. **Installation Mode**: Used for initial setup and calibration of seat movement limits.
2. **Manual Mode**: Direct control of seat position using physical buttons.
3. **Recall Mode**: Saving and retrieving customized seat positions from memory.

2.2.1 Installation Mode

Installation Mode is designed for initial setup and calibration of the seat control system. This mode runs an automatic calibration sequence that establishes the operational limits for all seat axes.

To Enter Installation Mode:

1. Press and hold all three RECALL buttons (RECALL_1, RECALL_2, and RECALL_3) simultaneously until all three motors of seat move backward (Approximately 3 seconds)
2. The system will begin the automatic installation sequence

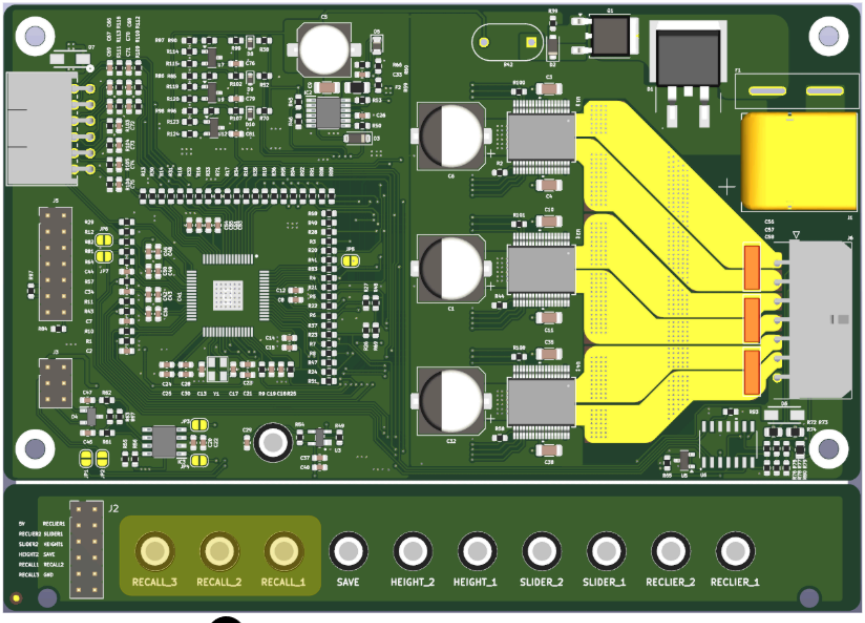




Figure 2.3: Activating Installation Mode by Pressing Three RECALL Buttons

Installation Sequence:

1. All three axes of the seat will move backward simultaneously to find their rear limits
2. Upon reaching the backward limits, the system resets the coordinate values for all three recall positions and recognizes this position as the "home" position
3. All three axes then move forward to find their forward limits
4. Upon reaching the forward limits, the system stores these positions as the maximum values for each axis
5. The system returns to IDLE state, ready for normal operation

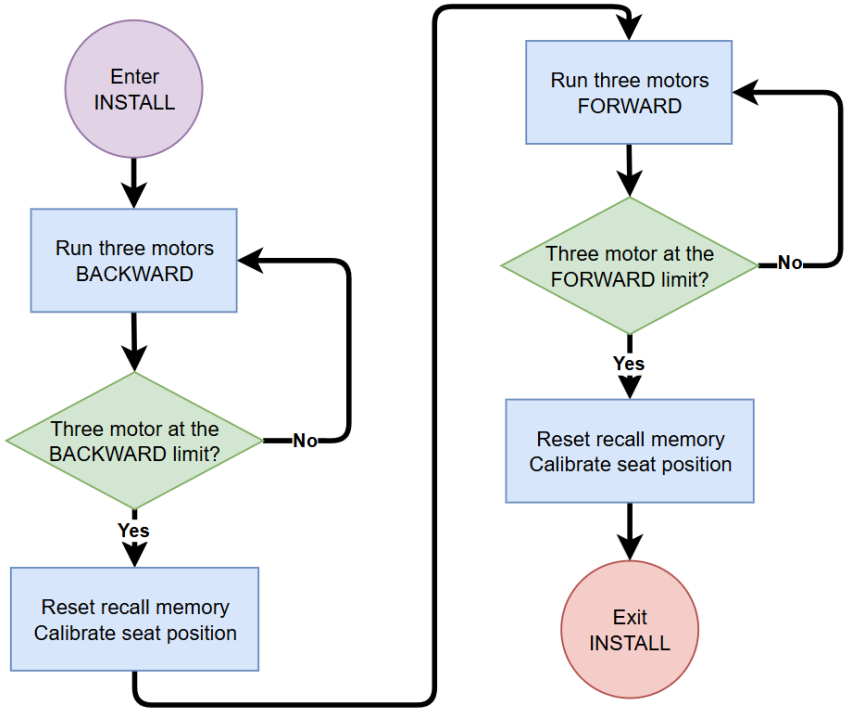


Figure 2.4: Installation Sequence Diagram

Note:

- Do not interrupt the installation process once it begins
- Ensure no obstacles are in the seat's path during calibration
- The entire process takes approximately 60-90 seconds to complete
- After completion, the seat will be ready for normal operation
- This process only needs to be performed once during initial installation or after maintenance

2.2.2 Manual Mode

Manual Mode is the standard operation mode for direct seat adjustment using the control buttons.

Manual Mode Operation:

- Press and hold any movement button (SLIDER_1, SLIDER_2, HEIGHT_1, HEIGHT_2, RECLINER_1, or RECLINER_2)
- The corresponding seat section will move in the indicated direction as long as the button is held
- Multiple buttons can be pressed simultaneously for compound movements
- Release buttons to stop movement

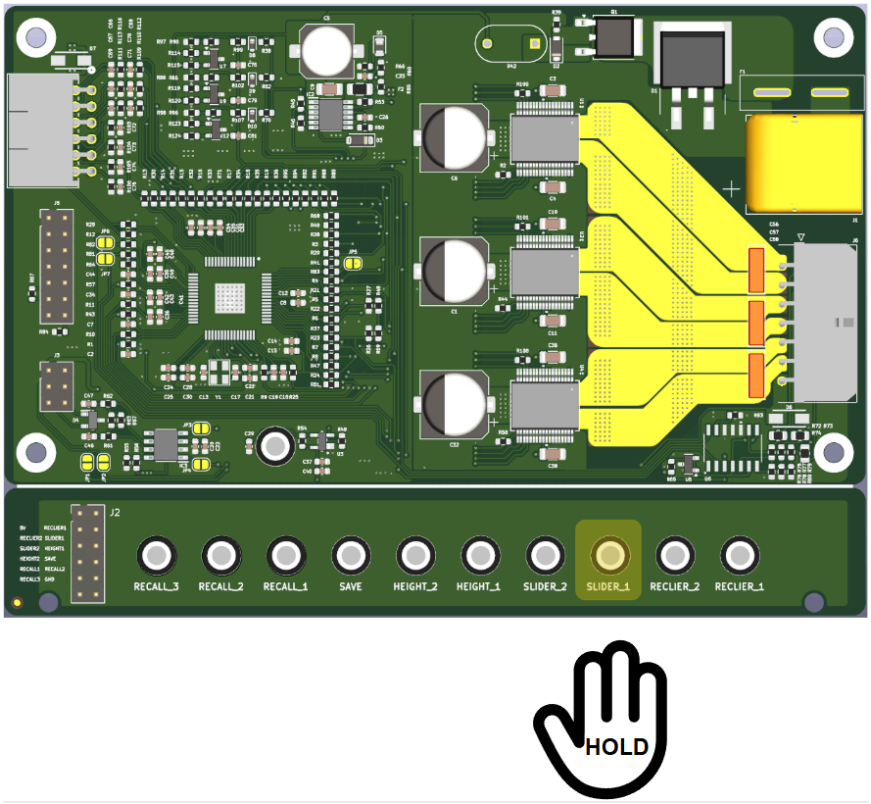


Figure 2.5: Manual Control Operation Example (Button SLIDER_1)

Example of Manual Operation:

To adjust the seat to a comfortable driving position:

1. Press and hold SLIDER_1 to move the seat forward
2. Release SLIDER_1 to stop seat at ideal position

Note:

- Movement stops immediately when button is released
- System automatically prevents movement beyond mechanical limits

2.2.3 Recall Mode

Recall Mode allows users to save and retrieve up to three customized seat positions. By default, all three recall position coordinates are initialized to 0, 0, 0, with each axis corresponding to recliner, slider, and height positions respectively.

Memory Functions:

The seat position is defined by coordinates along three axes:

- **Recliner axis**: Controls the angle of the seatback
- **Slider axis**: Controls forward/backward position of the entire seat
- **Height axis**: Controls the vertical position of the seat

To Save a Position:

1. Adjust the seat to your desired position using Manual Mode (all three axes)
2. Press the SAVE button once
3. Within 0.5 seconds, press one of the RECALL buttons (1, 2, or 3) to store the current position to that memory slot

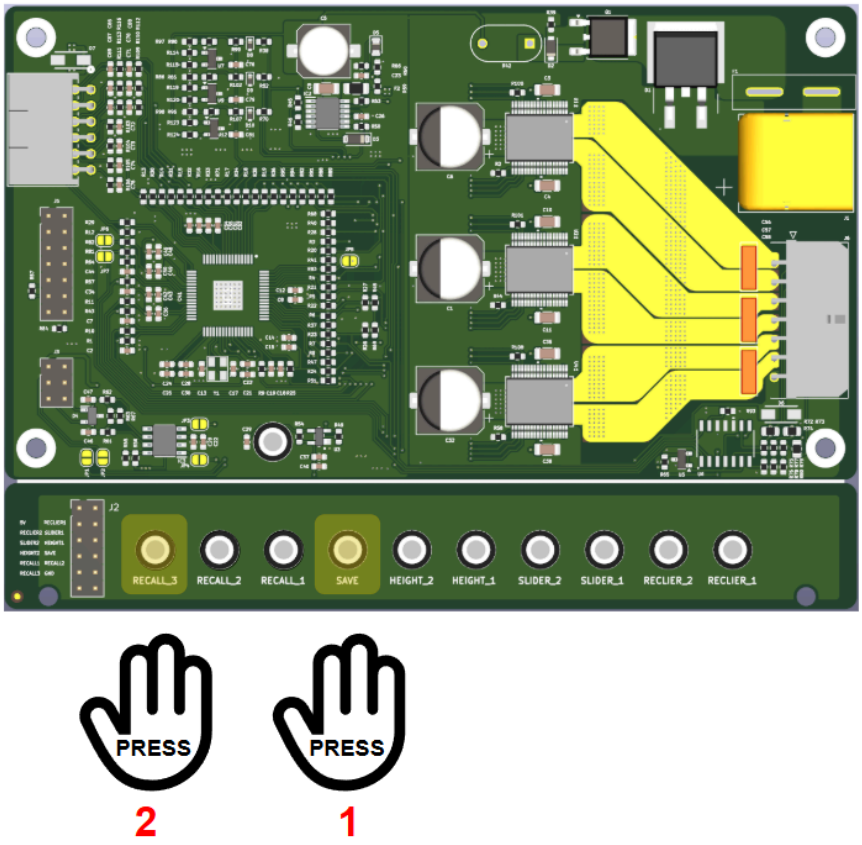


Figure 2.6: Saving a Seat Position to Memory

Example of Saving a Position:

To save your preferred driving position to memory slot 3:

1. Use manual controls to adjust all three axes (recliner, slider, height) to your ideal driving position
2. Press the SAVE button once (system enters save mode)
3. Press RECALL_3 within 0.5 seconds
4. Your preferred driving position is now stored in memory slot 3

To Recall a Position:

1. Press the corresponding RECALL button (1, 2, or 3) once
2. The seat will automatically move to the stored position, adjusting all three axes simultaneously
3. When the target position is reached, the system returns to IDLE state

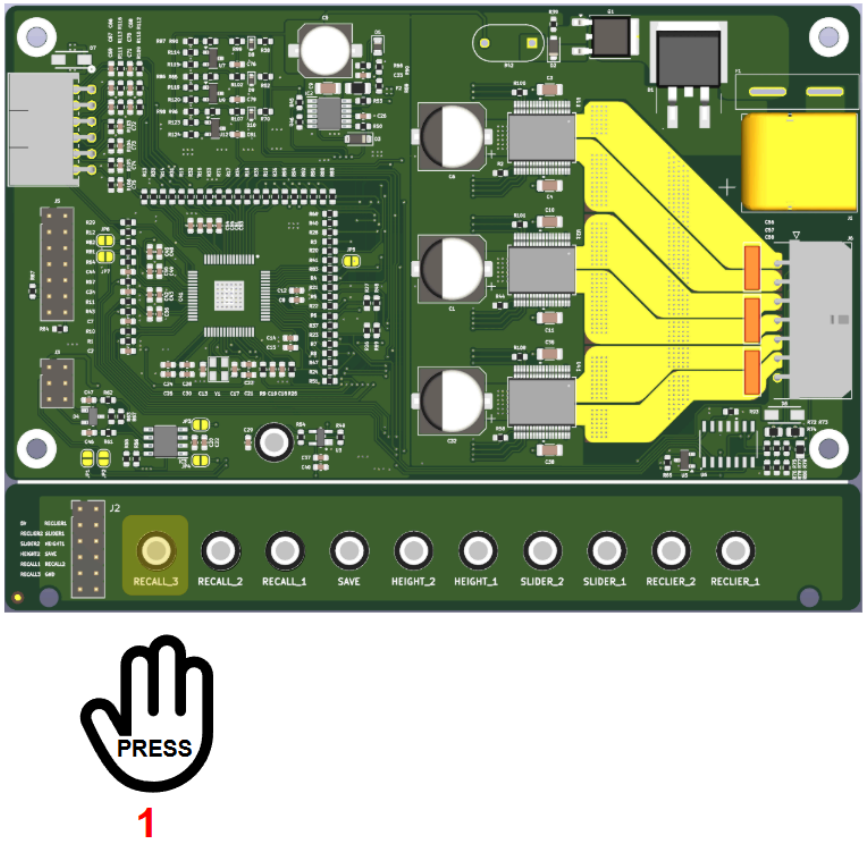


Figure 2.7: Recalling a Saved Position Using RECALL Button

Example of Using Recall Function:

To quickly return to your saved driving position after someone else has used the seat:

1. Enter the vehicle and press RECALL_3 once (assuming your position is saved in memory slot 3)
2. The seat will automatically move all three axes simultaneously to your saved position
3. Once the seat reaches the saved position, it will stop automatically and be ready for use

Note:

- Each user can have their own preferred positions stored in the three memory slots
- Saved positions are retained even when power is disconnected
- Positions can be overwritten at any time by repeating the save procedure

3. UART Communication Interface

3.1 Connection Setup

The Panthera system supports UART communication for control and diagnostic functions. Use the following parameters to establish connection:

Table 3.1: UART Connection Parameters

Parameter	Value	Description
Baud Rate	115200 bps	Communication speed
Data Bits	8	Bits per transmission
Stop Bits	1	Stop bit configuration
Parity	None	No parity bit
Flow Control	None	No hardware flow control

Physical Connection:

1. Connect GND of host device to system ground
2. Connect TX of host device to RX of Panthera system
3. Connect RX of host device to TX of Panthera system
4. Ensure power is supplied to the Panthera system

3.2 Command Structure

UART commands follow a structured format to ensure proper interpretation:

```
<START_BYTE><COMMAND_ID><DATA_LENGTH><DATA...><CHECKSUM><END_BYTE>
```

Where:

- `<START_BYTE>` : Fixed value 0x55 to mark command start
- `<COMMAND_ID>` : Single byte identifying the command type
- `<DATA_LENGTH>` : Length of data field in bytes (0-255)
- `<DATA...>` : Command-specific data (variable length)

- `<CHECKSUM>` : 8-bit sum of all previous bytes (modulo 256)
- `<END_BYTE>` : Fixed value 0xAA to mark command end

3.3 Basic Commands

Table 3.3: Basic UART Commands

Command ID	Description	Data Format
0x01	Get Status	No data required Response: System status byte
0x02	Move Motor	Byte 1: Motor ID (1-3) Byte 2: Direction (0=stop, 1=forward, 2=reverse) Byte 3: Speed (0-100%)
0x03	Stop All Motors	No data required
0x04	Get Position	Byte 1: Motor ID (1-3) Response: Current position value (2 bytes)
0x05	Set Absolute Position	Byte 1: Motor ID (1-3) Bytes 2-3: Target position
0x06	Save Position	Byte 1: Memory slot (1-3)
0x07	Recall Position	Byte 1: Memory slot (1-3)

Example: Move Motor Command

```
55 02 03 01 01 64 C0 AA
```

Breakdown:

- 0x55: Start byte
- 0x02: Command ID (Move Motor)
- 0x03: Data length (3 bytes)
- 0x01: Motor ID (Motor 1)
- 0x01: Direction (Forward)
- 0x64: Speed (100%)
- 0xC0: Checksum
- 0xAA: End byte

3.4 Advanced Commands

Command ID	Description	Data Format
0x10	Get System Diagnostics	No data required Response: Multiple bytes of system information
0x11	Get Motor Current	Byte 1: Motor ID (1-3) Response: Current value in mA (2 bytes)
0x12	Get Supply Voltage	No data required Response: Voltage in mV (2 bytes)
0x13	Get Error Log	Byte 1: Entry count (1-10) Response: Error codes with timestamps
0x14	Clear Error Log	No data required
0x20	Set Motor Parameters	Byte 1: Motor ID (1-3) Byte 2: Parameter ID Bytes 3+: Parameter value

3.5 Error Handling

The system responds to invalid commands with error messages:

Error Code	Description	Troubleshooting
0xE1	Invalid Command	Check command ID and resubmit
0xE2	Invalid Parameters	Verify parameter values are within acceptable range
0xE3	Checksum Error	Recalculate checksum and retry
0xE4	System Busy	Wait and retry command
0xE5	Motor Fault	Check motor status and clear faults before retrying
0xE6	Position Error	Verify target position is within valid range
0xE7	Communication Timeout	Check connection and retry

Error Response Format:

```
55 FF 02 <ERROR_CODE> <COMMAND_ID> <CHECKSUM> AA
```

Where:

- 0x55: Start byte
- 0xFF: Error response identifier
- 0x02: Data length
- `<ERROR_CODE>` : Specific error code from table above
- `<COMMAND_ID>` : ID of command that caused error
- `<CHECKSUM>` : Calculated checksum
- 0xAA: End byte

4. CAN Communication Interface

4.1 Hardware Configuration

The Panthera system supports standard CAN 2.0B communication with the following parameters:

Parameter	Value	Description
Baud Rate	500 kbps	CAN bus communication speed
Frame Format	Standard (11-bit)	CAN identifier format
Termination	120 Ohm	Required at both ends of CAN bus
Pin Assignment	CAN_H: Pin 50 CAN_L: Pin 51	Physical connection points

Connection Requirements:

1. Connect CAN_H of host to CAN_H of Panthera (Pin 50)
2. Connect CAN_L of host to CAN_L of Panthera (Pin 51)
3. Ensure proper termination at both ends of the CAN bus
4. Verify power and ground connections

4.2 Message Format

CAN messages use specific IDs to identify command types and target motors:

Message Type	CAN ID (hex)	Description
System Command	0x200	General system commands
Motor 1 Command	0x210	Commands for slider motor
Motor 2 Command	0x211	Commands for height motor
Motor 3 Command	0x212	Commands for recliner motor
System Status	0x300	General system status information
Motor 1 Status	0x310	Status data for slider motor
Motor 2 Status	0x311	Status data for height motor
Motor 3 Status	0x312	Status data for recliner motor

Message Type	CAN ID (hex)	Description
Diagnostic Data	0x400	Detailed diagnostic information

4.3 Control Commands

The following CAN messages can be used to control the Panthera system:

System Command (0x200)

Byte Position	Description	Values
0	Command Type	0x01: System reset 0x02: Stop all motors 0x03: Enter low-power mode 0x04: Exit low-power mode 0x05: Save position 0x06: Recall position
1	Parameter 1	Command-specific parameter
2	Parameter 2	Command-specific parameter
3-7	Reserved	Set to 0

Motor Command (0x210, 0x211, 0x212)

Byte Position	Description	Values
0	Command Type	0x01: Move to position 0x02: Move with speed 0x03: Stop motor 0x04: Reset encoder 0x05: Set parameters
1	Direction/Parameter Type	For move commands: 0: Stop 1: Forward 2: Reverse For parameter commands: Parameter ID
2-3	Position/Speed/Value	16-bit value for position or speed or parameter value
4-7	Additional Parameters	Command-specific additional parameters

Example: Move Motor 1 Forward at 75% Speed

```
ID: 0x210
Data: 02 01 4B 00 00 00 00 00
```

Breakdown:

- Byte 0: 0x02 (Move with speed command)
- Byte 1: 0x01 (Forward direction)
- Byte 2: 0x4B (75 decimal, representing 75% speed)
- Byte 3-7: 0x00 (Unused/reserved)

4.4 Status Monitoring

The system regularly broadcasts status messages that can be monitored:

System Status (0x300)

Byte Position	Description	Values
0	System State	0x00: IDLE 0x01: MANUAL 0x02: AUTO 0x03: SAVE 0x04: INIT 0x05: FAULT 0x06: CALIBRATING 0x07: SHUTDOWN
1	Error Flags	Bit 0: General Error Bit 1: Communication Error Bit 2: Motor 1 Error Bit 3: Motor 2 Error Bit 4: Motor 3 Error Bit 5: Supply Error Bit 6: Temperature Warning Bit 7: Reserved
2	Button Status 1	Bit 0: SLIDER_1 Bit 1: SLIDER_2 Bit 2: HEIGHT_1 Bit 3: HEIGHT_2 Bit 4: RECLINER_1 Bit 5: RECLINER_2 Bit 6-7: Reserved
3	Button Status 2	Bit 0: SAVE Bit 1: RECALL_1 Bit 2: RECALL_2 Bit 3: RECALL_3 Bit 4-7: Reserved
4-5	Supply Voltage	16-bit value in mV
6-7	Reserved	Set to 0

Motor Status (0x310, 0x311, 0x312)

Byte Position	Description	Values
0	Motor State	0x00: STOPPED 0x01: RUNNING_FORWARD 0x02: RUNNING_REVERSE 0x03: STALLED 0x04: FAULT
1	Motor Error Flags	Bit 0: Overcurrent Bit 1: Overtemperature Bit 2: Hall Sensor Error Bit 3: Encoder Error Bit 4: Position Error Bit 5: Driver Error Bit 6-7: Reserved
2-3	Current Position	16-bit position value
4-5	Current (mA)	16-bit current value in mA
6-7	Speed	16-bit speed value (0-100%)

4.5 Diagnostic Functions

The CAN interface provides advanced diagnostic capabilities:

Diagnostic Request (Host to Panthera)

- ID: 0x7DF (Standard Diagnostic Request ID)
- Length: 8 bytes
- Format: [Length, Service, PID, 0, 0, 0, 0, 0]

Diagnostic Response (Panthera to Host)

- ID: 0x7E8 (Standard Diagnostic Response ID)
- Length: 8 bytes
- Format: [Length, Service+0x40, PID, Data...]

Common Diagnostic PIDs:

PID	Description	Response Data
0x01	Request System Status	Multiple status flags
0x02	Request Error Codes	List of active error codes
0x03	Clear Error Codes	Success confirmation
0x04	Motor Parameters	Motor-specific parameters
0x05	System Information	Software version, hardware ID
0x06	Position Limits	Min/max positions for each motor
0x07	Operating Hours	System runtime statistics
0x08	Temperature Reading	System and motor temperatures

Example: Request System Information

```
ID: 0x7DF
Data: 02 09 05 00 00 00 00 00
```

Example Response:

```
ID: 0x7E8
Data: 07 49 05 01 23 45 67 89
```

Where:

- Byte 0: Length of data (7 bytes)
- Byte 1: Service response (0x49 = 0x09+0x40)
- Byte 2: PID (0x05 System Information)
- Bytes 3-7: System information data

5. Troubleshooting

5.1 Common Issues

Issue	Possible Causes	Troubleshooting Steps
No movement when buttons pressed	- Power supply issue - System in FAULT state - Button connection problem	1. Check power supply voltage 2. Verify system status via UART/CAN 3. Check button connections 4. Reset system if in fault state
Intermittent operation	- Loose connections - Thermal protection active - Current limiting	1. Check all electrical connections 2. Allow system to cool down 3. Check for obstructions 4. Verify voltage levels
No response to UART commands	- Incorrect baud rate - TX/RX reversed - Command format error	1. Verify baud rate settings 2. Check TX/RX connections 3. Verify command structure and checksum 4. Check power and ground connections
No response to CAN commands	- Incorrect baud rate - Missing termination - Incorrect message ID	1. Verify CAN bus configuration 2. Check termination resistors 3. Verify message IDs 4. Check physical connections
Memory positions not working	- Positions not saved - Memory corruption - Sensor failure	1. Re-save positions 2. Reset system 3. Check hall sensors 4. Verify motor operation

5.2 Diagnostic Procedures

System Reset Procedure:

1. Disconnect power for at least 10 seconds
2. Reconnect power
3. Allow system to initialize (approximately 5 seconds)
4. Test basic functionality

Button Test Procedure:

1. Enter diagnostic mode via UART command 0xD0

2. Press each button one at a time
3. System will report button status via UART
4. Exit diagnostic mode with command 0xD1

Motor Test Procedure:

1. Connect to system via UART or CAN
2. Issue motor movement commands at reduced speed (25%)
3. Verify smooth operation in both directions
4. Check current consumption during operation
5. Verify position reporting accuracy

Communication Test Procedure:

1. Send basic status request commands
2. Verify response format and data
3. Check for communication timeouts or errors
4. Test at various baud rates if issues persist